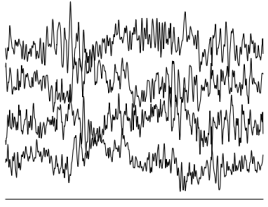
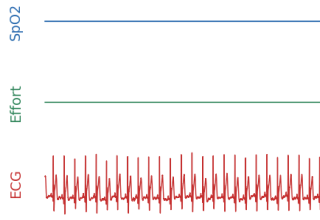


Two-stream multimodal multi-task network

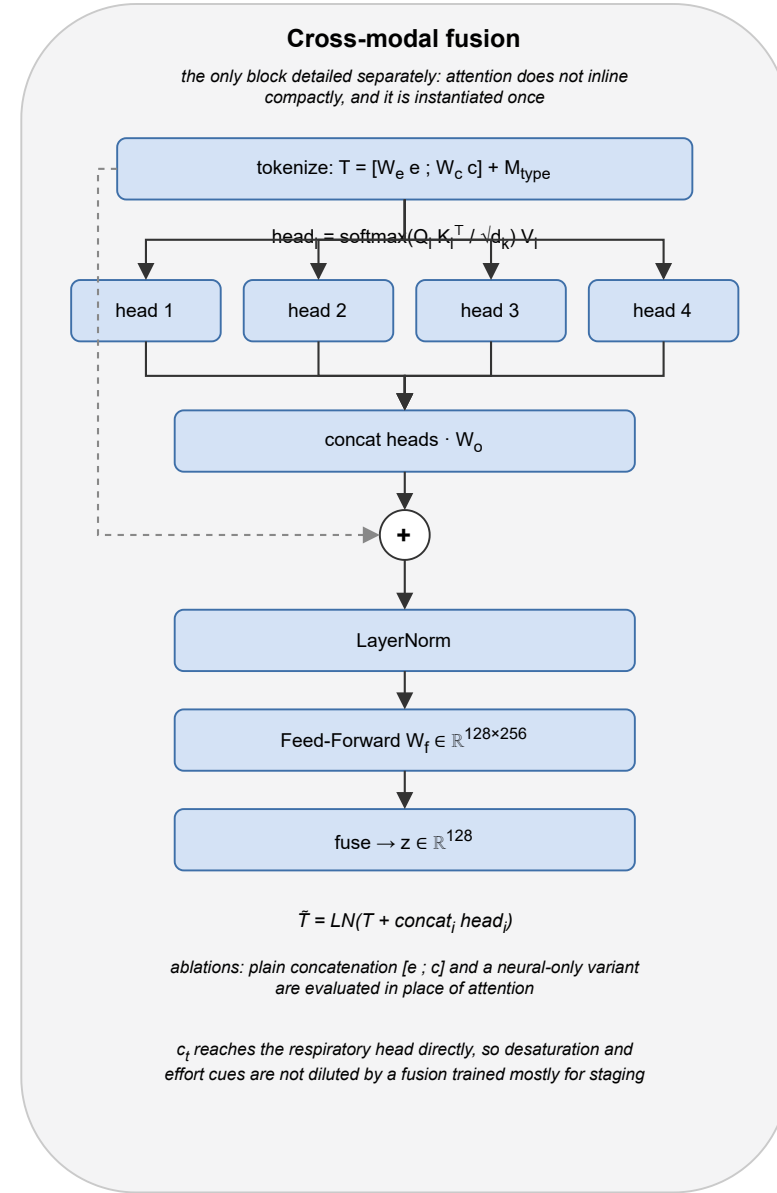
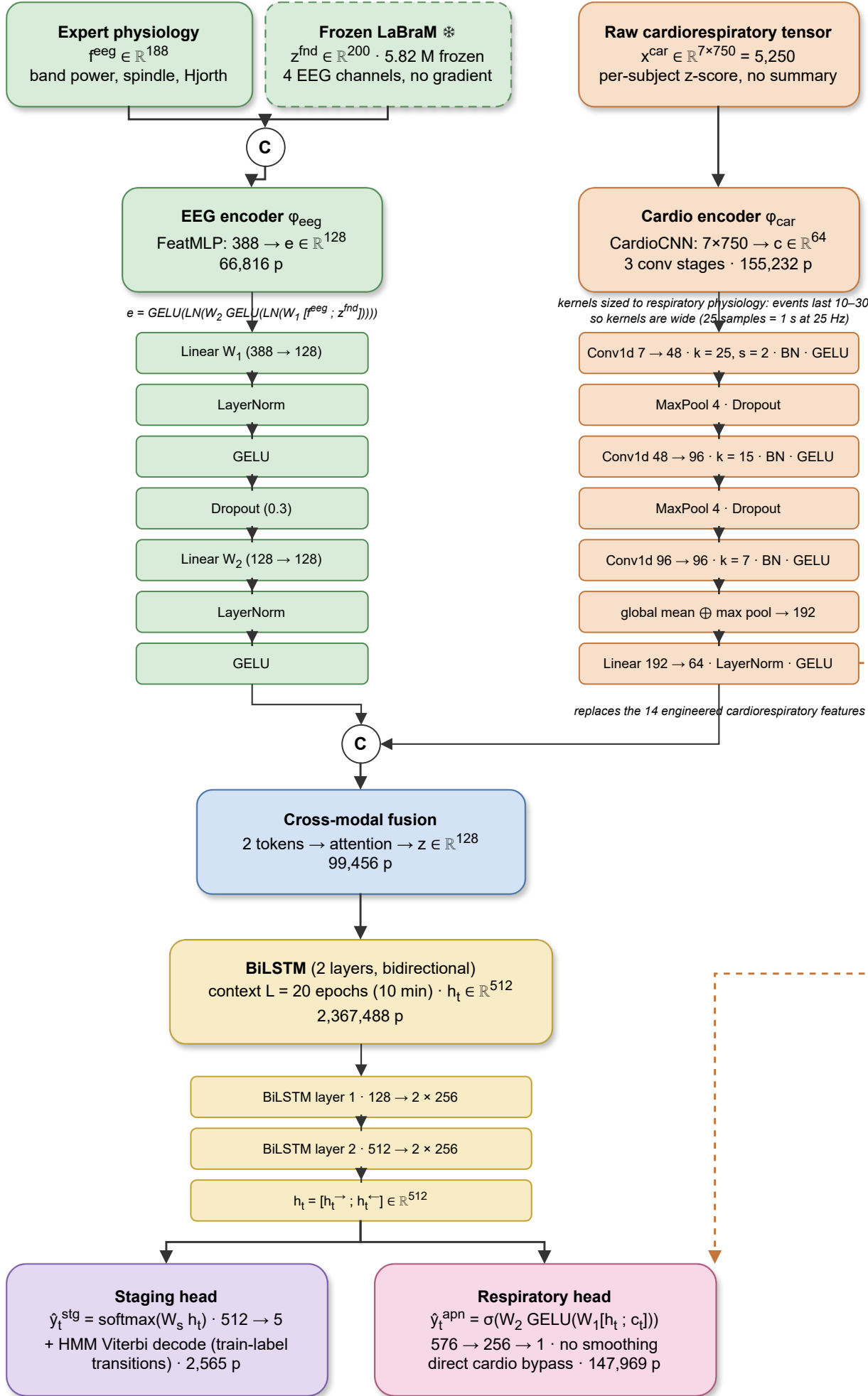
$$f_{\theta} : (f^{\text{eeg}}, z^{\text{fnd}}, x^{\text{car}}) \rightarrow (\hat{y}^{\text{stg}}, \hat{y}^{\text{apn}})$$



7-channel neural montage, 30 s @ 100 Hz



7-channel cardiorespiratory, 30 s @ 25 Hz



$$\text{Joint objective: } L = \text{CE}_{\sqrt{w}}(\hat{y}^{\text{stg}}, y^{\text{stg}}) + \lambda \text{BCE}_{\text{pw}}(\hat{y}^{\text{apn}}, y^{\text{apn}}), \lambda = 1$$

dashed outline = frozen (no gradient) $\cdot 2,764,774$ trainable + 5,819,936 frozen parameters